

Solutions - TD 05 (Arrays)

Exercise 01: Sum of Elements

Write an algorithm that reads an array of n integers and prints the sum of all elements.

Example:

Input: [2, 5, 3] → Output: 10

Pseudocode:

```
sum ← 0
for i ← 0 to n-1 do
    sum ← sum + A[i]
endfor
print sum
```

Exercise 02: Maximum Element

Find and print the largest value in an array.

Example:

Input: [4, 9, 1, 7] → Output: 9

Pseudocode:

```
max ← A[0]
for i ← 1 to n-1 do
    if A[i] > max then
        max ← A[i]
    endif
endfor
print max
```

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Exercise 03: Count Even Numbers

Count how many numbers in the array are even.

Example:

Input: [2, 5, 8, 11] → Output: 2

Pseudocode:

```
count ← 0  
  
for i ← 0 to n-1 do  
    if A[i] mod 2 = 0 then  
        count ← count + 1  
    endif  
endfor  
  
print count
```

Exercise 04: Reverse an Array

Print the array in reverse order.

Pseudocode:

```
for i ← n-1 downto 0 do  
    print A[i]  
endfor
```

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Exercise 05: Search for an Element x

Check if a given value x exists inside the array.

Example:

Array: [4, 6, 8], x = 6 → Output: "Found"

Pseudocode:

found $\leftarrow$ false

for i $\leftarrow$ 0 to n-1 do

 if A[i] = x then

 found $\leftarrow$ true

 endif

endfor

if found = true then

 print "Found"

else

 print "Not found"

endif

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Exercise 06: Average of Elements

Calculate the average (mean) of numbers in the array.

Example:

Input: [4, 6, 10] → Output: 6.66

Pseudocode:

```
sum ← 0
for i ← 0 to n-1 do
    sum ← sum + A[i]
endfor
average ← sum / n
print average
```

Exercise 07: Count Occurrences

Count how many times a number x appears in the array.

Example:

Array: [1, 2, 1, 3, 1], x=1 → Output: 3

Pseudocode:

```
count ← 0
for i ← 0 to n-1 do
    if A[i] = x then
        count ← count + 1
    endif
endfor
print count
```

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Exercise 08: Minimum Element

Find the minimum value in the array.

Example:

Input: [5, 3, 9, 1] → Output: 1

Pseudocode:

```
min ← A[0]
for i ← 1 to n-1 do
    if A[i] < min then
        min ← A[i]
    endif
endfor
print min
```

Exercise 09: Copy Array

Create a new array that is an exact copy of the original one.

Pseudocode:

```
for i ← 0 to n-1 do
    B[i] ← A[i]
endfor
print B
```

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Exercise 10: Swap First and Last Elements

Swap the first element with the last element of the array.

Example:

Input: [7, 4, 9] → Output: [9, 4, 7]

Pseudocode:

temp $\leftarrow$ A[0]

A[0] $\leftarrow$ A[n-1]

A[n-1] $\leftarrow$ temp

print A